Ai Assisted Coding

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Batch:24

Experiment 6: AI-Based Code Completion: Working with suggestions for classes, loops, conditionals

Task1: Use an AI code completion tool to generate a loop-based program.

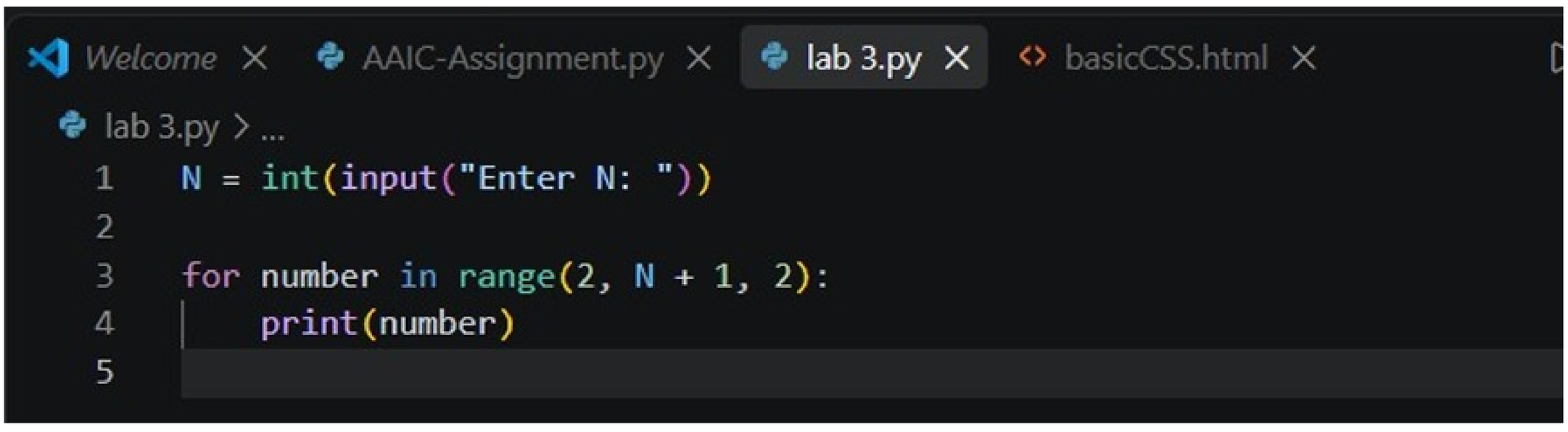
Prompt:

"Generate Python code to print all even numbers between 1 and N using a loop."

Expected Output:

- AI-generated loop logic.
- Identification of loop type used (for or while).
- Validation with sample inputs.

Input



Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & C:\Users\ASUS\AppData\Local\Prog
rams\Python\Python313\python.exe "c:/Users/ASUS/OneDrive/Desktop/AAIC-LAB/lab
3.py"
Enter N: 10
2
4
6
8
10
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & C:\Users\ASUS\AppData\Local\Prog
rams\Python\Python313\python.exe "c:/Users/ASUS/OneDrive/Desktop/AAIC-LAB/lab
3.py"
Enter N: 25
2
4
6
8
10
12
14
16
18
20
22
24
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB>
```

Task2: Use an AI code completion tool to combine loops and conditionals.

Prompt:

"Generate Python code to count how many numbers in a list are even and odd."

Expected Output:

- AI-generated code using loop and if condition.
- Correct count validation.
- Explanation of logic flow.

Input

```
lab 3.py > ...
numbers = list(map(int, input("Enter numbers separated by spaces: ").spl

even_count = sum(number % 2 == 0 for number in numbers)
odd_count = sum(number % 2 != 0 for number in numbers)

print("Even numbers:", even_count)
print("Odd numbers:", odd_count)

```

Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & C:\Users\ASUS\AppData\Local\Prog
rams\Python\Python313\python.exe "c:/Users/ASUS/OneDrive/Desktop/AAIC-LAB/lab
3.py"
Enter numbers separated by spaces:  10 15 22 33 48 51 60 7
Even numbers: 4
Odd numbers: 4
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & C:\Users\ASUS\AppData\Local\Prog
rams\Python\Python313\python.exe "c:/Users/ASUS/OneDrive/Desktop/AAIC-LAB/lab
3.py"
Enter numbers separated by spaces: 1 14 10 96 02 45
Even numbers: 4
Odd numbers: 2
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB>
```

Task3: Use an AI tool to complete a Python class that validates user input.

Prompt:

"Generate a Python class User that validates age and email using

conditional statements."

Expected Output:

- AI-generated class with validation logic.
- Verification of condition handling.
- Test cases for valid and invalid inputs.

Input

lab3.py

```python
class User:
    def __init__(self, name, age, email):
        self.name = name
        self.age = age
        self.email = email
        self.is_valid = self.validate()

    def validate(self):
        if self.age < 0 or self.age > 120:
            print("Invalid age: age must be between 0 and 120")
            return False
        if "@" not in self.email or "." not in self.email:
            print("Invalid email: must contain '@' and '.'")
            return False
        print(f"{self.name} is valid. Age: {self.age}, Email: {self.email}")
        return True


name = input("Enter name: ")
age = int(input("Enter age: "))
email = input("Enter email: ")
user = User(name, age, email)
```

Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & python lab3.py
Enter name: Vinay
Enter age: 25
Enter email: vinay@gmail.com
Vinay is valid. Age: 25, Email: vinay@gmail.com
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & python lab3.py
Enter name: Sam
Enter age: 150
Enter email: sam@gmail.com
Invalid age: age must be between 0 and 120
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & python lab3.py
Enter name: Anu
Enter age: 30
Enter email: anugmail.com
Invalid email: must contain '@' and '.'
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB>
```

Task4: Use an AI code completion tool to generate a Python class for

managing student details.

Prompt:

"Generate a Python class Student with attributes (name, roll number,

marks) and methods to calculate total and average marks."

Expected Output:

- AI-generated class code.
- Verification of correctness and completeness of class

structure.

- Minor manual improvements (if needed) with justification.

Input

```python
class Student:
    def __init__(self, name, roll_number, marks):
        self.name = name
        self.roll_number = roll_number
        self.marks = marks

    def total_marks(self):
        return sum(self.marks)

    def average_marks(self):
        return self.total_marks() / len(self.marks)


name = input("Enter student name: ")
roll_number = input("Enter roll number: ")
marks = list(map(int, input("Enter marks separated by spaces: ").split()))

student = Student(name, roll_number, marks)
print(f"Total marks: {student.total_marks()}")
print(f"Average marks: {student.average_marks():.2f}")
```

Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & python lab3.py
Enter student name: Rahul
Enter roll number: 2503A52710
Enter marks separated by spaces: 85 90 78 92 88
Total marks: 433
Average marks: 86.60
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB>
```

Task5: Use an AI tool to generate a complete Python program using classes, loops, and conditionals together.

Prompt:

"Generate a Python program for a simple bank account system using
class, loops, and conditional statements."

Expected Output:

- Complete AI-generated program.
- Identification of strengths and limitations of AI suggestions.
- Reflection on how AI assisted coding productivity.

-

Input

```python
class BankAccount:
    def __init__(self, owner, balance=0):
        self.owner = owner
        self.balance = balance

    def deposit(self, amount):
        if amount <= 0:
            print("Deposit amount must be positive.")
        else:
            self.balance += amount
            print(f"Deposited {amount}. New balance: {self.balance}")

    def withdraw(self, amount):
        if amount <= 0:
            print("Withdrawal amount must be positive.")
        elif amount > self.balance:
            print("Insufficient balance.")
        else:
            self.balance -= amount
            print(f"Withdrew {amount}. New balance: {self.balance}")

    def check_balance(self):
        print(f"Current balance: {self.balance}")


name = input("Enter account holder name: ")
account = BankAccount(name)

while True:
    print("1. Deposit  2. Withdraw  3. Check Balance  4. Exit")
    choice = input("Enter choice: ")
    if choice == "1":
        amount = float(input("Enter amount to deposit: "))
        account.deposit(amount)
    elif choice == "2":
        amount = float(input("Enter amount to withdraw: "))
        account.withdraw(amount)
    elif choice == "3":
        account.check_balance()
    elif choice == "4":
        print("Exiting... Thank you!")
        break
    else:
        print("Invalid choice. Please try again.")
```

Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & python lab3.py
Enter account holder name: Priya
1. Deposit  2. Withdraw  3. Check Balance  4. Exit
Enter choice: 1
Enter amount to deposit: 5000
Deposited 5000.0. New balance: 5000.0
1. Deposit  2. Withdraw  3. Check Balance  4. Exit
Enter choice: 3
Current balance: 5000.0
1. Deposit  2. Withdraw  3. Check Balance  4. Exit
Enter choice: 2
Enter amount to withdraw: 7000
Insufficient balance.
1. Deposit  2. Withdraw  3. Check Balance  4. Exit
Enter choice: 2
Enter amount to withdraw: 2000
Withdrew 2000.0. New balance: 3000.0
1. Deposit  2. Withdraw  3. Check Balance  4. Exit
Enter choice: 4
Exiting... Thank you!
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB>
```